INCH-POUND

MIL-PRF-83726/12G
15 January 2024
SUPERSEDING
MIL-R-83726/12F
7 January 2016

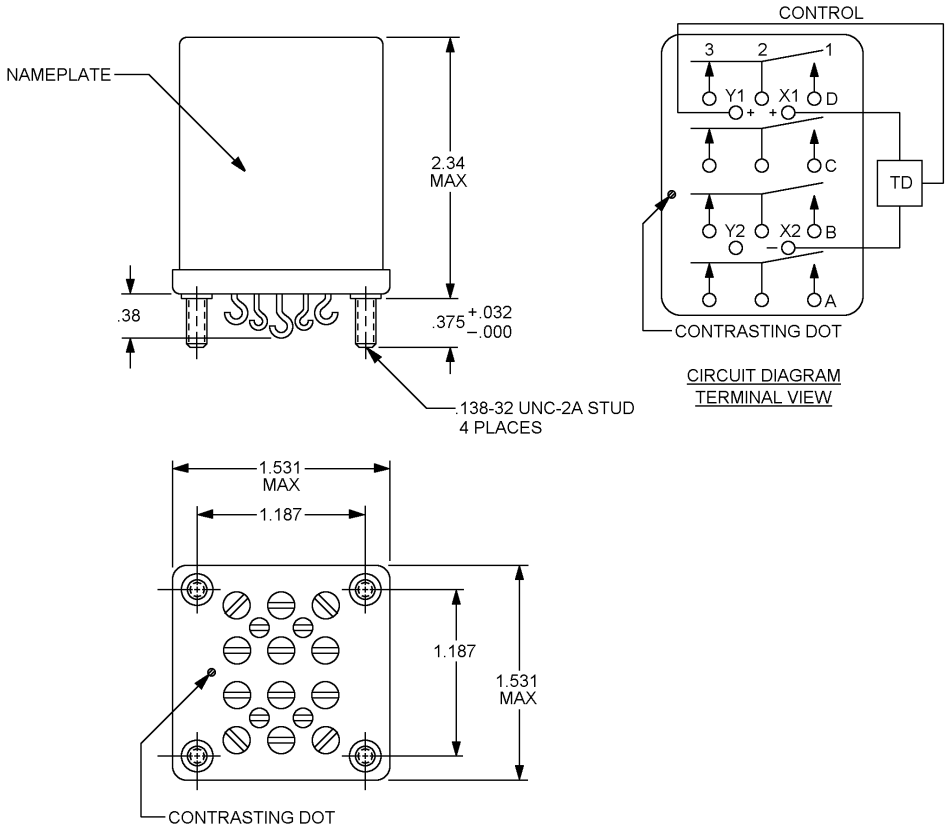
PEROFRMANCE SPECIFICATION SHEET

# RELAYS, HYBRID, TIME DELAY (ON RELEASE), TYPE IIA, CLASS B, 10 AMPERES, 4PDT, HERMETICALLY SEALED, FIXED TIME, 0.100 TO 300 SECONDS

Inactive for new design after 10 January 1994.
For new design use MIL-PRF-83726/29.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-83726.




| Inches | mm |
|---|---|
| .032 | 0.81 |
| .138 | 3.51 |
| .375 | 9.53 |
| .38 | 9.7 |
| 1.187 | 30.15 |
| 1.531 | 38.89 |
| 2.34 | 59.4 |

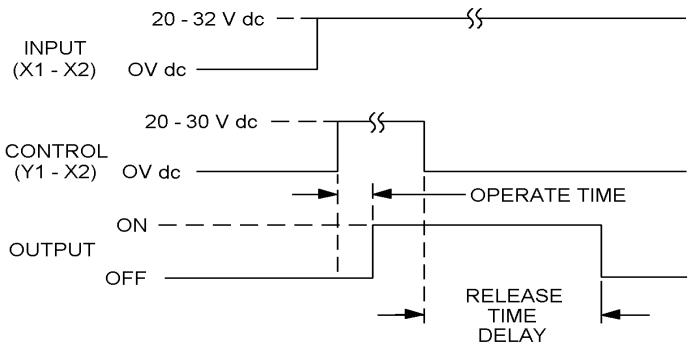
FIGURE 1. Outline dimensions and configuration of relay.

AMSC N/A

FSC 5945

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TIMING DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm) for three place decimals and ±.03
3. (0.8 mm) for two place decimals.
4. Metric equivalents are given for general information only.
5. Terminal numbers shall not appear on the relay header. There shall be affixed to the relay a permanent legible schematic diagram that identifies each terminal location specified.

FIGURE 1. Outline dimensions and configuration of relay - Continued.

TABLE I. Dash numbers and time delay characteristics. 1/

| Dash number | Time delay (seconds) ±10% | Dash number | Time delay (seconds) ±10% | Dash number | Time delay (seconds) ±10% | Dash number | Time delay (seconds) ±10% |
|---|---|---|---|---|---|---|---|
| | | -1202 | 12 | -6502 | 65 | -1703 | 170 |
| -1000 | 2/ .100 | -1402 | 14 | -7002 | 70 | -1803 | 180 |
| -5000 | 2/ .500 | -1602 | 16 | -7502 | 75 | -1903 | 190 |
| -7500 | 2/ .750 | -1802 | 18 | -8002 | 80 | -2003 | 200 |
| -1001 | 1 | -2002 | 20 | -8502 | 85 | -2103 | 210 |
| -2001 | 2 | -2202 | 22 | -9002 | 90 | -2203 | 220 |
| -3001 | 3 | -2502 | 25 | -9502 | 95 | -2303 | 230 |
| -4001 | 4 | -3002 | 30 | -1003 | 100 | -2403 | 240 |
| -5001 | 5 | -3502 | 35 | -1103 | 110 | -2503 | 250 |
| -6001 | 6 | -4002 | 40 | -1203 | 120 | -2603 | 260 |
| -7001 | 7 | -4502 | 45 | -1303 | 130 | -2703 | 270 |
| -8001 | 8 | -5002 | 50 | -1403 | 140 | -2803 | 280 |
| -9001 | 9 | -5502 | 55 | -1503 | 150 | -2903 | 290 |
| -1002 | 10 | -6002 | 60 | -1603 | 160 | -3003 | 300 |

1/ Additional time delay relays within the .100 to 300 second delay range are available. To establish Part or Identifying Numbers (PIN's) not listed in table I (see "PIN" herein).
2/ Add ±10 milliseconds to ±10 percent tolerance.



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REQUIREMENTS:

Contact data:

Configuration: 4PDT.

Life/load ratings (relay case grounded):

| Type of load | Life (cycles) | 28 V dc | Amperes 115/200 1 and 3 phase 400 Hz |
|---|---|---|---|
| Resistive | 100,000 | 10 | 10 |
| Inductive | 20,000 | 8 | 8 |
| Motor | 100,000 | 4 | 4 |
| Lamp | 100,000 | 2 | 2 |
| Reduced current resistive | 400,000 | 2.5 | 2.5 |

Contact voltage drop:

Initial: 0.150 volt.

Intermediate current: Applicable in accordance with MIL-PRF-83536.

Contact bounce: 1 millisecond maximum.

Overload:

DC: 40 amperes.

AC: 60 amperes.

Rupture:

DC: 50 amperes.

AC: 80 amperes.

Input data:

Duty rating: Continuous.

Maximum voltage: 30 V dc.

Nominal voltage (over temperature range): 28 V dc.

Minimum voltage: 20 V dc.

Minimum voltage high temperature test: 21 V dc.



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Minimum voltage continuous current test: 23.5 V dc.

Maximum current at 25°C: 0.150 ampere.

Release time delay: See table I.

Control current: 20 milliamperes maximum.

Control pulse duration: 50 milliseconds minimum.

Operate time: 50 milliseconds maximum.

Electrical data:

Insulation resistance at 500 V dc: 1/

Initial: 1,000 megohms minimum.

After life or environmental tests: 500 megohms minimum.

Dielectric strength (sea level):

| | Initial | After life tests |
|---|---|---|
| Input (X1 - X2 - Y1) 1/ | 1,000 V rms | 1,000 V rms |
| All other points | 1,250 V rms | 1,000 V rms |

Dielectric strength (altitude): 80,000 feet. 1/

Input (X1 - X2 - Y1): 350 V rms. 1/

All other points: 350 V rms.

Transients: In accordance with MIL-STD-704 for 28 V dc system (figure 11).

1/ Input terminals X1, X2, and Y1 shall be connected together during this test.



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Environmental data:

Temperature range (operating): -55°C to +85°C.

Maximum altitude rating: 80,000 feet.

Shock g-level: 100 g's, 1/2 sine, 3 axes.

Duration: 6 ±1 milliseconds.

Maximum duration contact opening: 10 microseconds.

Vibration (sinusoidal):

G-level: 20 g's.

Frequency range: 10 Hz to 3,000 Hz.

Vibration (random): Applicable in accordance with MIL-STD-202- 214, test condition 1B.

Power spectral density: 0.4 $g^2$/Hz.

Frequency range: 50 Hz to 2,000 Hz.

Duration: 15 minutes each plane.

Acceleration: 20 g's in any axis.

Seal: Not applicable.

Humidity: 95 percent relative humidity.

Physical data:

Configuration and dimensions: See figure 1.

Terminations: Solder hook.

Terminal strength: 3 ±0.5 pounds pull.

Weight: 10 ounces (280 grams) maximum.

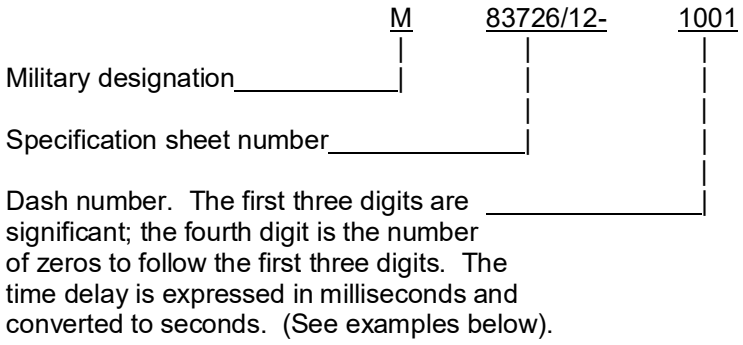
Marking: In accordance with MIL-PRF-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

Electrostatic Discharge Sensitive (ESDS) control program: Applicable (see MIL-PRF-83726).



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Part or Identifying Number (PIN): Consists of the prefix M83726/12-, and a four-digit dash number (expressed in milliseconds) as follows:




Examples: M83726/12-1000 - 100 millisecond time delay.
M83726/12-1001 - 1 second time delay.

CONFORMANCE INSPECTION:

Performance of groups B and C tests are not applicable.

The Qualified Products List (QPL) associated with this inactive for new design specification will be maintained until acquisition of the product is no longer required, whereupon this specification and the QPL will be canceled.



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NOTE: As of 15 June 1999, MIL-PRF-83726 no longer specifies Quality Levels, but existing order configurations may still include them. Relays with a “W” quality level indicator at the end are interchangeable with relays without a quality level letter.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-83726, this document references the following:

MIL-PRF-83726/29 MIL-PRF-83536 MIL-STD-202-214 MIL-STD-704
MIL-STD-1285

Custodians:
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2024-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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